Standard Model

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Abstract

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1 Gauge Redundancy

In order to describe a relativistic QFT containing massless vector particles, we use a gauge theory. For a vector field A_μ, the familiar redundancy is

$$A_\mu \to A_\mu + \partial_\mu \alpha, \tag{1}$$

where $\alpha(x)$ is an arbitrary function. We now derive the origin of this redundancy from one-particle representations of the Poincare group.

1.1 One-particle representations

A relativistic one-particle state furnishes a unitary representation of the Poincare group. A Poincare transformation acts as

$$x'^\mu = \Lambda^\mu_\nu x^\nu + a^\mu, \quad \Lambda \in \mathrm{SO}(3,1). \tag{2}$$

Writing $g = g(\Lambda, a)$, the group law is

$$g(\Lambda_1, a_1) g(\Lambda_2, a_2) = g(\Lambda_1 \Lambda_2, \Lambda_1 a_2 + a_1), \tag{3}$$

and hence

$$\begin{aligned} g^{-1}(\Lambda, a) &= g(\Lambda^{-1}, -\Lambda^{-1} a), \\ g(\Lambda, a) &= g(1, a) g(\Lambda, 0). \end{aligned} \tag{4}$$

Thus we are able to separate the translation and Lorentz transformation parts and each part can be written in terms of generators and exp maps. Translations and Lorentz transformations are generated by

$$g(1, a) = \exp\big(i P^\mu a_\mu\big), \quad g(\Lambda, 0) = \exp\Big(-\frac{i}{2} \omega_{\mu\nu} J^{\mu\nu}\Big). \tag{5}$$

The algebra of the generators is:

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(\eta^{\mu\sigma} J^{\nu\rho} + \eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\sigma} J^{\mu\rho}), \tag{6}$$

$$[P^\mu, J^{\rho\sigma}] = i(\eta^{\mu\rho} P^\sigma - \eta^{\mu\sigma} P^\rho), \quad [P^\mu, P^\nu] = 0. \tag{7}$$

Since the momentum components commute, we label one-particle states by their momentum and by an internal label σ:

$$P^\mu \ |p, \sigma\rangle = p^\mu \ |p, \sigma\rangle. \tag{8}$$

Under translation they transforms as a phase factor and under a Lorentz transformation,

$$U(\Lambda) \ |p, \sigma\rangle = \sum_{\sigma'} C_{\sigma\sigma'}(\Lambda, p) \ |\Lambda p, \sigma'\rangle. \tag{9}$$

Lorentz transformations preserves the following quantities of a state (this means that the LT matrix commute with these operators):

$$p^2 = m^2, \quad W^2, \quad \text{sign}(p^0), \tag{10}$$

where W^μ is the Pauli-Lubanski vector. This gives us an approach to classify the representations of the Lorentz group cause states with different values of the conserved quantities cannot be connected by a Lorentz transformation.

First of all, the orbits of the Lorentz group in momentum space are classified by the mass m and the sign of p^0, physically we don't consider negative energy and 0 energy correspond to vaccum, the left over orbits are classified with different mass. For each orbit, choose a representative momentum k^μ. For every p on the same orbit, choose a standard Lorentz transformation $L(p)$ satisfying

$$p^\mu = L^\mu_\nu(p) k^\nu. \tag{11}$$

Then define

$$|p, \sigma\rangle := N_p U(L(p)) \ |k, \sigma\rangle. \tag{12}$$

where we leave σ as the index of some unknown parameters and N_p is a normalization factor. Under a general Lorentz transformation,

$$\begin{aligned} U(\Lambda)|p, \sigma\rangle &= N_p U(\Lambda L(p))|k, \sigma\rangle \\ &= N_p U(L(\Lambda p)) U(L^{-1}(\Lambda p)\Lambda L(p))|k, \sigma\rangle. \end{aligned} \tag{13}$$

The transformation

$$W(\Lambda, p) := L^{-1}(\Lambda p)\Lambda L(p) \tag{14}$$

leaves k invariant:

$$W^\mu_\nu k^\nu = k^\mu. \tag{15}$$

It therefore belongs to the **Wigner's little group** of k. Its action can be defined as some representation of the little group on the internal label σ:

$$U(W)|k, \sigma\rangle = \sum_{\sigma'} D_{\sigma'\sigma}(W)|k, \sigma'\rangle, \tag{16}$$

Consequently,

$$U(\Lambda) \ |p, \sigma\rangle = \frac{N_p}{N_{\Lambda p}} \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p)) \ |\Lambda p, \sigma'\rangle. \tag{17}$$

Thus, classifying one-particle states reduces to classifying representations of the little group.

1.1.1 Massive particles

For $m > 0$, choose

$$k^\mu = (m, 0, 0, 0). \tag{18}$$

Spatial rotations leave k^μ invariant, so the little group is SO(3) (or its double cover SU(2)). Its finite-dimensional irreducible representations are labeled by

$$s = 0, \frac{1}{2}, 1, \frac{3}{2}, ... \tag{19}$$

> **Remark**
>
> In fact SO(3) doesn't have spin half representation, but SU(2) does. The little group is actually SU(2) which is the double cover of SO(3). This may relate to the fact that our symmetry group is not really SO(3,1).

1.1.2 Massless particles

For $m = 0$, choose

$$k^\mu = E(1, 0, 0, 1). \tag{20}$$

The little group is generated by rotations about the z-axis J^3 and two generators

$$T_1 := K^1 - J^2, \quad T_2 := K^2 + J^1, \tag{21}$$

with

$$J^i := \frac{1}{2}\varepsilon^{ijk}J^{jk}, \quad K^i := J^{0i}. \quad \varepsilon^{123} = 1. \tag{22}$$

In the four-vector representation, we can easily show that the generators generate the Wigner's little group of k:

$$(J^{\mu\nu})^\rho_\sigma = i(\delta^\nu_\sigma \eta^{\mu\rho} - \delta^\mu_\sigma \eta^{\nu\rho}). \tag{23}$$

For the chosen k, one finds

$$\begin{aligned} K^1 k = J^2 k = iE(0, 1, 0, 0), \\ K^2 k = -J^1 k = iE(0, 0, 1, 0), \end{aligned} \tag{24}$$

and therefore $T_1 k = T_2 k = 0$.

Their algebra is

$$[J^3, T_1] = iT_2, \quad [J^3, T_2] = -iT_1, \quad [T_1, T_2] = 0. \tag{25}$$

This is the Lie algebra of ISO(2): J^3 generates rotations in the plane, while T_1 and T_2 generate translations. Because T_1 and T_2 commute, choose simultaneous eigenstates to label the states in the representation:

$$T_1 \ |a, b\rangle = a \ |a, b\rangle, \quad T_2 \ |a, b\rangle = b \ |a, b\rangle. \tag{26}$$

Using the Baker-Campbell-Hausdorff formula,

$$\begin{aligned} e^{i\theta J^3} T_1 e^{-i\theta J^3} &= T_1 \cos\theta - T_2 \sin\theta, \\ e^{i\theta J^3} T_2 e^{-i\theta J^3} &= T_2 \cos\theta + T_1 \sin\theta. \end{aligned} \tag{27}$$

Therefore

$$e^{-i\theta J^3} \ |a, b\rangle = |a\cos\theta - b\sin\theta, a\sin\theta + b\cos\theta\rangle. \tag{28}$$

Thus we can build a representation of the little group by taking a circle of states with the same radius in the (a, b) plane. The radius

$$\rho := \sqrt{a^2 + b^2} \tag{29}$$

is invariant. If $\rho \neq 0$, the representation contains a continuous circle of states and is infinite-dimensional. An equivalent way of showing that we have infinite number of state is by defining

$$T_+ := T_1 + iT_2, \quad T_- := T_1 - iT_2, \tag{30}$$

this gives

$$[J^3, T_+] = T_+, \quad [J^3, T_-] = -T_-, \quad [T_+, T_-] = 0. \tag{31}$$

If

$$J^3|h\rangle = h|h\rangle, \tag{32}$$

then

$$\begin{aligned} J^3 T_+|h\rangle &= (h+1)T_+|h\rangle, \\ J^3 T_-|h\rangle &= (h-1)T_-|h\rangle. \end{aligned} \tag{33}$$

Unlike ordinary angular momentum, this ladder does not terminate when $\rho \neq 0$. Such representations are the continuous-spin representations which is not observed in the physical world. For the familiar finite-helicity particles, we impose the trivial action

$$J^3|p,h\rangle = h|p,h\rangle, \quad T_i|p,h\rangle = 0. \tag{34}$$

The pure translation part of the little group may be represented by

$$W^\mu{}_\nu(\alpha,\beta) = \begin{pmatrix} 1+\zeta & \alpha & \beta & -\zeta \\ \alpha & 1 & 0 & -\alpha \\ \beta & 0 & 1 & -\beta \\ \zeta & \alpha & \beta & 1-\zeta \end{pmatrix}, \quad \zeta \coloneqq \frac{\alpha^2+\beta^2}{2}. \tag{35}$$

Indeed,

$$W^\mu{}_\nu k^\nu = k^\mu. \tag{36}$$

Though we wish the translation part of the little group to act trivially on the quantum state, yet if we define a spin 1 theory that have local degrees of freedom that transforms as a 4 vector, then it must have polarization internal degrees of freedom that transforms non-trivially under the translation part of the little group. To resolve this, we need to identify different polarization vectors and understand them as the same physical state. This is the origin of gauge redundancy.

Take the two transverse polarization vectors (we care about transverse polarization because the EoMs of massless vector field implies that the polarization vector must be transverse to the momentum vector)

$$\varepsilon_1^\mu = (0,1,0,0), \quad \varepsilon_2^\mu = (0,0,1,0). \tag{37}$$

Then

$$W\varepsilon_1 = \begin{pmatrix} \alpha \\ 1 \\ 0 \\ \alpha \end{pmatrix} = \varepsilon_1 + \frac{\alpha}{E}k, \quad W\varepsilon_2 = \begin{pmatrix} \beta \\ 0 \\ 1 \\ \beta \end{pmatrix} = \varepsilon_2 + \frac{\beta}{E}k. \tag{38}$$

The shifted polarization remains transverse and has the same norm:

$$k\cdot(\varepsilon + ck) = k\cdot\varepsilon = 0, \quad (\varepsilon+ck)^2 = \varepsilon^2, \tag{39}$$

because $k^2 = 0$. We impose that the polarization vectors

$$\varepsilon^\mu \to \varepsilon^\mu + ck^\mu \tag{40}$$

must describe the same physical state.

1.2 Two-component Spinors

1.2.1 Spinor Representations

We have discussed the representation of Lorentz group on the Hilbert space. Now we focus on the representation of Lorentz group on the space of fields. Considering the standard model, we mainly care about the spin 1/2 representations. The six Lorentz generators

$$J^i := \frac{1}{2}\varepsilon^{ijk}J^{jk}, \quad K^i := J^{0i} \tag{41}$$

satisfy

$$\begin{aligned} [J^i, J^j] &= i\varepsilon^{ijk}J^k, \\ [J^i, K^j] &= i\varepsilon^{ijk}K^k, \\ [K^i, K^j] &= -i\varepsilon^{ijk}J^k. \end{aligned} \tag{42}$$

Define the complex combinations

$$N_i^+ := \frac{1}{2}(J^i + iK^i), \quad N_i^- := \frac{1}{2}(J^i - iK^i). \tag{43}$$

They obey

$$\begin{aligned} \left[N_i^+, N_j^+\right] &= i\varepsilon_{ijk}N_k^+, \\ \left[N_i^-, N_j^-\right] &= i\varepsilon_{ijk}N_k^-, \\ \left[N_i^+, N_j^-\right] &= 0. \end{aligned} \tag{44}$$

Thus the complexified Lorentz algebra is two commuting copies of $\mathrm{su}(2)$, and finite-dimensional irreducible representations are labeled by (j_L, j_R).

- For the undotted $\left(\frac{1}{2}, 0\right)$ representation,

$$J^i = \frac{\sigma^i}{2}, \quad K^i = -i\frac{\sigma^i}{2}. \tag{45}$$

A rotation by $\boldsymbol{\theta}$ and boost of rapidity $\boldsymbol{\eta}$ are represented by

$$\Lambda = \exp\left(i(\boldsymbol{\theta} - i\boldsymbol{\eta}) \cdot \frac{\boldsymbol{\sigma}}{2}\right) \in \mathrm{SL}(2, \mathbb{C}), \quad \det\Lambda = 1. \tag{46}$$

An undotted spinor transforms as

$$\lambda_\alpha \to \Lambda_\alpha{}^\beta \lambda_\beta. \tag{47}$$

> **Remark**
>
> By convention the LT matrix have the first index as the row index as lower index and the second index as the column index as upper index. Always stick to this convention.

- For the dotted $\left(0, \frac{1}{2}\right)$ representation,

$$J^i = \frac{\sigma^i}{2}, \quad K^i = +i\frac{\sigma^i}{2}, \tag{48}$$

and the conjugate spinor transforms as

$$\tilde{\lambda}_{\dot{\alpha}} \to (\Lambda^*)_{\dot{\alpha}}{}^{\dot{\beta}} \tilde{\lambda}_{\dot{\beta}} = \tilde{\lambda}_{\dot{\beta}} \left(\Lambda^\dagger\right)^{\dot{\beta}}{}_{\dot{\alpha}} \tag{49}$$

Remark

One important observation is that the complex conjugate of a (1/2, 0) representation is a (0, 1/2) representation. Thus, the dot and undotted indices can be viewed as related to each other with a CC.

$$\lambda_\alpha^* = \tilde{\lambda}_{\dot{\alpha}}, \tag{50}$$

The lorentz transformation of the (0,1/2) representation can also be written as the complex conjugate of the (1/2,0) representation as above:

$$\tilde{\lambda}_{\dot{\alpha}} \to (\Lambda^*)_{\dot{\alpha}}{}^{\dot{\beta}} \tilde{\lambda}_{\dot{\beta}} \tag{51}$$

Here we also take the convention that the first index is the row index and the second index is the column index, we only rename them as dotted indices to show that they live in the (0,1/2) representation space.

Now we have the basic conventions for the spinors and spinor indices. We can further define raising and lowering of spinor indices to define LT invariant contractions. Dotted and undotted indices are raised and lowered with antisymmetric tensors:

$$\lambda^\alpha = \varepsilon^{\alpha\beta}\lambda_\beta, \quad \lambda_\alpha = \varepsilon_{\alpha\beta}\lambda^\beta, \quad \varepsilon^{\alpha\beta}\varepsilon_{\beta\gamma} = \delta^\alpha_\gamma, \quad \varepsilon^{12} = 1, \quad \varepsilon_{12} = -1. \tag{52}$$

$$\tilde{\lambda}^{\dot{\alpha}} = \varepsilon^{\dot{\alpha}\dot{\beta}}\tilde{\lambda}_{\dot{\beta}}, \quad \tilde{\lambda}_{\dot{\alpha}} = \varepsilon_{\dot{\alpha}\dot{\beta}}\tilde{\lambda}^{\dot{\beta}}, \quad \varepsilon^{\dot{\alpha}\dot{\beta}}\varepsilon_{\dot{\beta}\dot{\gamma}} = \delta^{\dot{\alpha}}_{\dot{\gamma}}, \quad \varepsilon^{\dot{1}\dot{2}} = 1, \quad \varepsilon_{\dot{1}\dot{2}} = -1. \tag{53}$$

Under a Lorentz transformation, the four types of spinors transform as

$$\lambda_\alpha \to \Lambda_\alpha{}^\beta \lambda_\beta, \quad \lambda^\alpha \to (\Lambda^{-1})_\beta{}^\alpha \lambda^\beta, \quad \tilde{\lambda}_{\dot{\alpha}} \to (\Lambda^*)_{\dot{\alpha}}{}^{\dot{\beta}} \tilde{\lambda}_{\dot{\beta}}, \quad \tilde{\lambda}^{\dot{\alpha}} \to \left((\Lambda^*)^{-1}\right)_{\dot{\beta}}{}^{\dot{\alpha}} \tilde{\lambda}^{\dot{\beta}}. \tag{54}$$

Remark

To consist with the following conventions, it is better to use $\Lambda^\dagger$ instead of Λ^*. (just take a transpose) Thus we can write the transformations of the dotted spinors as:

$$\tilde{\lambda}_{\dot{\alpha}} \to \tilde{\lambda}_{\dot{\beta}} {\Lambda^\dagger}^{\dot{\beta}}{}_{\dot{\alpha}}, \quad \tilde{\lambda}^{\dot{\alpha}} \to \left(\Lambda^\dagger\right)^{-1\,\dot{\alpha}}{}_{\dot{\beta}} \tilde{\lambda}^{\dot{\beta}}. \tag{55}$$

now the upper index dotted spinor behaves like a ket.

We can thus see that the upper and lower indices transform in the opposite way, and the dotted and undotted indices transform in the complex conjugate way. The inverse property may give us a chance of defining a Lorentz invariant contraction of spinors. We first introduce a braket notation for the spinors, such that the contraction of vectors gives a LT invariant scalar. The spinor-helicity bra-ket notation is

$$\langle\lambda| \coloneqq \lambda^\alpha, \quad |\lambda\rangle \coloneqq \lambda_\alpha, \quad [\lambda| \coloneqq \tilde{\lambda}_{\dot{\alpha}}, \quad |\lambda] \coloneqq \tilde{\lambda}^{\dot{\alpha}}. \tag{56}$$

The two Lorentz-invariant contractions are

$$\langle pq\rangle \coloneqq p^\alpha q_\alpha = \varepsilon^{\alpha\beta} p_\beta q_\alpha, \tag{57}$$

$$[pq] \coloneqq \tilde{p}_{\dot{\alpha}} \tilde{q}^{\dot{\alpha}}. \tag{58}$$

> **Remark**
>
> One should be careful about the ordering of the contraction. In the $\langle\rangle$ contraction, the first one have upper index and the second one have lower index. In the [] contraction, the first one have lower index and the second one have upper index. This is in general important due to the fact that lower and upper ε are not the same and both antisymmetric.

Both are antisymmetric:

$$\langle pq\rangle = -\langle qp\rangle, \quad [pq] = -[qp], \quad \langle pp\rangle = [pp] = 0. \tag{59}$$

1.2.2 Momentum as a bispinor

Indeed the spin 1/2 field can be taken as a 2 component spinor field. Moreover, we can also represent 4 vectors as composite spinors. A four-momentum can be mapped to a 2×2 matrix:

$$p_{\alpha\dot{\alpha}} \coloneqq p_\mu \sigma^\mu_{\alpha\dot{\alpha}} = \begin{pmatrix} p^0 - p^3 & -p^1 + ip^2 \\ -p^1 - ip^2 & p^0 + p^3 \end{pmatrix}. \tag{60}$$

Defining

$$p^+ \coloneqq p^0 + p^3, \quad p^- \coloneqq p^0 - p^3, \quad p_\perp \coloneqq p^1 + ip^2, \tag{61}$$

we may also write

$$p_{\alpha\dot{\alpha}} = \begin{pmatrix} p^- & -p_\perp^* \\ -p_\perp & p^+ \end{pmatrix}. \tag{62}$$

If we do a Lorentz transformation to the 4 momentum, the bispinor transforms as the spinors with suitable index dot and position: (that's why this is called a bispinor not meerly a matrix)

$$p_{\alpha\dot{\alpha}} \to \Lambda_\alpha{}^\beta p_{\beta\dot{\beta}} \Lambda^{\dagger\dot{\beta}}{}_{\dot{\alpha}}. \tag{63}$$

> **Remark**
>
> Notice that the dotted and undotted and upper and lower indices are defined to suite the convention of lorentz transformation. The index always transforms as 2 component spinors with the same position and doted or undotted.

Its determinant is the Lorentz invariant

$$\det p_{\alpha\dot{\alpha}} = (p^0)^2 - (p^1)^2 - (p^2)^2 - (p^3)^2 = p^2 = m^2. \tag{64}$$

Therefore

- for a massive momentum, $\det p \neq 0$ and $\mathrm{rank}(p) = 2$;

$$p^2 \neq 0: \quad \det(p_{\alpha\dot{\alpha}}) \neq 0, \quad \mathrm{rank}(p_{\alpha\dot{\alpha}}) = 2, \tag{65}$$

- for a massless momentum, $\det p = 0$ and $\mathrm{rank}(p) = 1$.

$$p^2 = 0: \quad \det(p_{\alpha\dot{\alpha}}) = 0, \quad \mathrm{rank}(p_{\alpha\dot{\alpha}}) = 1. \tag{66}$$

Remark

By convention, we write $\sigma^{\mu}{}_{\alpha\dot{\alpha}}$ and $\overline{\sigma}^{\mu\dot{\alpha}\alpha}$. They are related by

$$\overline{\sigma}^{\mu\dot{\alpha}\alpha} = \varepsilon^{\alpha\beta}\varepsilon^{\dot{\alpha}\dot{\beta}}\sigma^{\mu}{}_{\beta\dot{\beta}}. \tag{67}$$

When raising or lowering both indices, their order is exchanged. This is to suit the convention of the matrix and the first index is always the row index and the second is the column index. Thus the momentum matrix under raising of indices is:

$$p^{\dot{\alpha}\alpha} = p_{\mu}\overline{\sigma}^{\mu\dot{\alpha}\alpha} = \begin{pmatrix} p^0 + p^3 & p^1 - ip^2 \\ p^1 + ip^2 & p^0 - p^3 \end{pmatrix}. \tag{68}$$

1.3 Spinors for Massless Particles

1.3.1 Momentum and Basis Spinors

In the massless case, the rank-one matrix factorizes:

$$p_{\alpha\dot{\alpha}} = \lambda_{\alpha}\tilde{\lambda}_{\dot{\alpha}} = |p\rangle[p|. \tag{69}$$

Choose the reference momentum

$$k^{\mu} = E(1, 0, 0, 1). \tag{70}$$

With the convention $k_{\alpha\dot{\alpha}} = k_{\mu}\sigma^{\mu}_{\alpha\dot{\alpha}}$ and $k^{\dot{\alpha}\alpha} = k_{\mu}\overline{\sigma}^{\mu\dot{\alpha}\alpha}$, we have:

$$k^{\dot{\alpha}\alpha} = 2E\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = k^{\dot{\alpha}}k^{\alpha}. \tag{71}$$

Under a Lorentz transformation, the dot on the index and the position of the index

$$p^{\dot{\alpha}\alpha} = \Lambda^{\dagger\,\dot{\alpha}}{}_{\dot{\beta}}k^{\dot{\beta}\beta}\Lambda_{\beta}{}^{\alpha} = p^{\dot{\alpha}}p^{\alpha}. \tag{72}$$

For a real null momentum of energy E and direction (θ, φ):

$$p^{\mu} = E(1, \sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta), \tag{73}$$

one may choose the following spinors to represent the momentum:

$$|p] = \sqrt{2E}\begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right)e^{-i\varphi} \end{pmatrix}, \quad |p\rangle = \sqrt{2E}\begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right)e^{i\varphi} \end{pmatrix}. \tag{74}$$

remember the difference between bras and kets are raising and lowering index and the difference between $|p\rangle$ and $|p]$ is complex conjugation and a raising and lowering of index. The remaining spinors are obtained by raising and lowering indices:

$$p_{\dot{\alpha}} = \varepsilon_{\dot{\alpha}\dot{\beta}}p^{\dot{\beta}}, \quad p^{\alpha} = \left(p^{\dot{\alpha}}\right)^{*}, \quad p_{\alpha} = \varepsilon_{\alpha\beta}p^{\beta}. \tag{75}$$

A little-group rotation around the direction of p acts as

$$|p\rangle \to e^{i\frac{\phi}{2}}|p\rangle, \quad |p] \to e^{-i\frac{\phi}{2}}|p], \tag{76}$$

and leaves $p_{\alpha\dot{\alpha}} = |p\rangle[p|$ invariant. Thus, everything is consistent. We say that the spinor $|p\rangle$ has helicity $-\frac{1}{2}$ and the spinor $|p]$ has helicity $+\frac{1}{2}$, from now on we will use helicity weight to denote how a quantity transform under rotation around a certain direction (most probabaly the momentum direction).

Remark

For a real null momentum, $|p\rangle$ and $|p]$ are defined only up to opposite phase factors:

$$|p\rangle \to e^{i\phi}|p\rangle, \quad |p] \to e^{-i\phi}|p]. \tag{77}$$

This is the action of the rotational SO(2) subgroup of the massless little group ISO(2), corresponding to rotations around the direction of p.

One now may also realize that the spinors constructed here can natually be the wave function of spin 1/2 particles (i.e. when Weyl fermion act on vaccum gives a spinor wave function) in the helicity basis. The result is that:

Remark

Here we use the 4 component spinors notation for convenience in calculation. Though it can understand as 2 Weyl spinors combined together.

$$\begin{aligned} u_-(p) &= |p\rangle, \quad u_+(p) = |\, p], \quad v_+(p) = |p\rangle, \quad v_-(p) = |\, p], \\ \overline{u}_-(p) &= [p\,|, \quad \overline{u}_+(p) = \langle p|, \quad \overline{v}_+(p) = [p\,|, \quad \overline{v}_-(p) = \langle p|. \end{aligned} \tag{78}$$

Here we abuse the notation to use 2 component spinors to represent 4 component spinors. What we really mean is that:

$$u_-(p) = \begin{pmatrix} |p\rangle \\ 0 \end{pmatrix}, \quad u_+(p) = \begin{pmatrix} 0 \\ |p] \end{pmatrix}, \tag{79}$$

$$v_+(p) = \begin{pmatrix} |p\rangle \\ 0 \end{pmatrix}, \quad v_-(p) = \begin{pmatrix} 0 \\ |p] \end{pmatrix}. \tag{80}$$

$$\overline{u}_-(p) = (0, [p\,|), \quad \overline{u}_+(p) = (\langle p|, 0), \tag{81}$$

$$\overline{v}_+(p) = (0, [p\,|), \quad \overline{v}_-(p) = (\langle p|, 0). \tag{82}$$

The $|\, p\rangle$ ket always stays on the upper of left of the 4 component spinor and the $|\, p]$ ket always stays on the lower or right of the 4 component spinor. Another commonly used notation is that:

$$p_\mu \gamma^\mu = |p\rangle[p| + |p]\langle p|. \tag{83}$$

All this series of notation makes it convenience to calculate diagrams containing fermions.

1.3.2 Polarization Vectors

Now we turn to the polarization vectors of the massless vector fields. We know that under Wigner's little group the polarization vectors transform nontrivially. Thus it need more than the momentum spinors to construct the polarization vectors. We introduce an auxiliary null reference momentum ξ such that

$$\xi^2 = 0, \quad \langle p\xi\rangle \neq 0, \quad [p\xi] \neq 0. \tag{84}$$

$$\varepsilon^+_{\alpha\dot{\alpha}}(p, \xi) = \sqrt{2}\frac{1}{\langle \xi p\rangle}|\xi\rangle[p|, \tag{85}$$

$$\varepsilon^-_{\alpha\dot{\alpha}}(p, \xi) = \sqrt{2}\frac{1}{[p\xi]}(|p\rangle[\xi|). \tag{86}$$

One can see that they satisfy

$$p \cdot \varepsilon^{\pm} = 0, \quad \varepsilon^{+} \cdot \varepsilon^{-} = -1. \tag{87}$$

Moreover, under little group rotations they transform nontrivially:

$$\varepsilon^{\pm} \to e^{\pm i\phi}\varepsilon^{\pm}. \tag{88}$$

This is consistent with the fact that the polarization vectors are transverse to the momentum and have helicity ± 1.

The reference spinor is arbitrary. Since $\langle p\xi\rangle \neq 0$, another choice can be written as

$$|\xi'\rangle = a|\xi\rangle + b|p\rangle. \tag{89}$$

Then

$$\varepsilon^{+}_{\alpha\dot{\alpha}}(p,\xi') = \varepsilon^{+}_{\alpha\dot{\alpha}}(p,\xi) + \frac{\sqrt{2}b}{a\langle\xi p\rangle}p_{\alpha\dot{\alpha}}. \tag{90}$$

Thus changing the reference spinor shifts the polarization by a term proportional to the momentum. The requirement that observables be independent of ξ is the gauge redundancy.

For sake of calculation, we have another expression of the polarization vectors in terms of the 4 compnent spinors and the gamma matrices:

$$\varepsilon^{\mu}_{-}(p;q) = \frac{\langle p|\gamma^{\mu}|q]}{\sqrt{2}[qp]}, \quad \varepsilon^{\mu}_{+}(p;q) = \frac{\langle q|\gamma^{\mu}|p]}{\sqrt{2}\langle qp\rangle}. \tag{91}$$

1.4 Spinors for massive particles

1.4.1 Momentum and Basis Spinors

For a massive momentum,

$$\det(p_{\alpha\dot{\alpha}}) = p^2 = m^2 \neq 0, \tag{92}$$

so the rank-two bispinor can be decomposed into two rank-one matrices:

$$p_{\alpha\dot{\alpha}} = l^1_{\alpha\dot{\alpha}} + l^2_{\alpha\dot{\alpha}} = |1\rangle[1| + |2\rangle[2|, \quad \left(l^1\right)^2 = \left(l^2\right)^2 = 0. \tag{93}$$

This decomposition is not unique. In the rest frame,

$$k^{\mu} = (m,0,0,0), \quad l^{\mu}_1 = \frac{m}{2}(1,\boldsymbol{n}), \quad l^{\mu}_2 = \frac{m}{2}(1,-\boldsymbol{n}). \tag{94}$$

The two null rays define the spin-quantization axis (just a stupid name for now)

$$n^{\mu} = \frac{1}{m}(l^{\mu}_1 - l^{\mu}_2), \quad n^2 = -1, \quad n \cdot k = 0. \tag{95}$$

> **Remark**
>
> This axis is not unique, we have a freedom of choosing an internal direction. One can understand as we use two massless particles moving in opposite directions to construct a massive particle momentum.

Under a rotation by an angle ϕ around $\boldsymbol{n}$, the two null momenta l_1 and l_2 remain invariant. Using the spinor transformation rules, we find

$$|1] \to e^{-i\frac{\phi}{2}}\,|1], \quad |1\rangle \to e^{i\frac{\phi}{2}}|1\rangle, \quad |2] \to e^{i\frac{\phi}{2}}\,|2], \quad |2\rangle \to e^{-i\frac{\phi}{2}}|2\rangle. \tag{96}$$

Thus we say that the spinors have the helicity of:

- $|1]$: $h = +\frac{1}{2}$ $|1\rangle$: $h = -\frac{1}{2}$ $|2]$: $h = -\frac{1}{2}$ $|2\rangle$: $h = +\frac{1}{2}$

We may notice that the helicity of 1 and 2 are opposite, this is due to they in fact are moving in opposite directions. We take the direction of l_1 as the spin-quantization axis, thus the spinors associated with l_2 have opposite helicity.

> **Remark**
>
> Notice that this is not true helicity, cause it is just the "spin" along a chosen axis. But we split the massless momentum using this axis as the momentum direction, thus we call it helicity.

Before we notice that the momentum spinors naturally give us the "wavefunction" (sorry for not having a better phrase) of the Weyl fermions in helicity basis. Now we can also see that the massive momentum spinors naturally give us the "wavefunction" of the Dirac fermions in helicity basis. They are:

$$u_+(k) = \begin{pmatrix} |2\rangle \\ |1] \end{pmatrix}, \quad u_-(k) = \begin{pmatrix} |1\rangle \\ |2] \end{pmatrix}. \tag{97}$$

One can prove that they solve the Dirac equation in the rest frame. Both Weyl components describe the same "helicity" state. In fact, this relation can be generalized to all momentum and we will discuss about it below.

> **Remark**
>
> One should keep in mind that helicity is not well defined for massive particles. Nevertheless, here we means helicity along some hand chosen axis where the massive particle is combined by two massless particles.

In the rest frame,

$$k_{\alpha\dot{\alpha}} = m \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{98}$$

A rotation leaves k invariant, however it rotates the two null momenta l_1 and l_2:

$$RkR^\dagger = k, \quad Rl^iR^\dagger \neq l^i. \tag{99}$$

Thus the massive little group acts nontrivially on the two null momenta and rotates the spin-quantization axis.

For a general momentum, we can decompose it into the following two parts:

$$p^\mu = l^\mu + \frac{m^2}{2l \cdot q} q^\mu, \quad l^2 = q^2 = 0. \tag{100}$$

We can find that $p^2 = m^2$. A convenient spinor decomposition is

$$|1\rangle = |l\rangle, \quad |2\rangle = \frac{m}{\langle ql \rangle} |q\rangle, \quad |1] = |l], \quad |2] = \frac{m}{[lq]} |q], \tag{101}$$

where we use a useful identity is for any null l and q,

$$2l \cdot q = \langle lq \rangle [ql]. \tag{102}$$

Moreover, we can also combine the two spinor together and amazingly find a way to solve the dirac equation for a general momentum and get the wavefunctions of the Dirac fermions:

$$u_+(p) = \begin{pmatrix} \frac{m}{\langle lq\rangle}|q\rangle \\ |l] \end{pmatrix}, \quad u_-(p) = \begin{pmatrix} |l\rangle \\ \frac{m}{[lq]}|q] \end{pmatrix}. \tag{103}$$

> **Remark**
>
> With suitable choice, we can recover the standard wavefunction in Helicity basis. For generic q choice, we get different basis, using direction of q to define spin up and spin down.

Now we consider the massless limit. For any fixed q,

$$m^2 \to 0 \quad \Rightarrow \quad p^\mu \to l^\mu. \tag{104}$$

The wavefunctions reduce to

$$u_+(p) = \begin{pmatrix} \frac{m}{\langle lq\rangle}|q\rangle \\ |l] \end{pmatrix} \to \begin{pmatrix} 0 \\ |l] \end{pmatrix}, \quad u_-(p) = \begin{pmatrix} |l\rangle \\ \frac{m}{[lq]}|q] \end{pmatrix} \to \begin{pmatrix} |l\rangle \\ 0 \end{pmatrix}. \tag{105}$$

This is a result we are familiar with which is massless Dirac spinor has a single nonvanishing chirality.

1.4.2 Polarization Vectors

The massive-vector polarization bispinors are

$$\varepsilon^+ = \frac{|1]\langle 2|}{\sqrt{2}m}, \quad \varepsilon^- = \frac{|2]\langle 1|}{\sqrt{2}m}, \quad \varepsilon^0 = \frac{|1]\langle 1| - |2]\langle 2|}{m}. \tag{106}$$

They have the correct little-group transformation and obey

$$p \cdot \varepsilon = 0. \tag{107}$$

We now take the massless limit. transverse polarizations remain finite:

$$\varepsilon^+ \to \frac{|\ l]\langle q\ |}{\sqrt{2}\langle lq\rangle} \quad \varepsilon^- \to \frac{|\ l\rangle[q\ |}{\sqrt{2}[lq]}. \tag{108}$$

We can see q now becomes the massless reference vector. A normalized longitudinal polarization can be written as

$$\varepsilon_0^\mu(p;q) = \frac{1}{m}\left(l^\mu - \frac{m^2}{2l\cdot q}q^\mu\right) = \frac{p^\mu}{m} - \frac{m}{p\cdot q}q^\mu. \tag{109}$$

Its massless limit is singular, but the divergent part is proportional to p^μ and is therefore pure gauge.

1.4.3 From polarizations to fields

In QFT, we play with fields. The vector field can be expanded in the following terms in terms of the polarization vector.

$$A_\mu(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3 2E_p} \sum_{h=\pm1} \left(\varepsilon_\mu^h(p)a_h(p)e^{-ip\cdot x} + \varepsilon_\mu^{h*}(p)a_h^\dagger(p)e^{ip\cdot x}\right). \tag{110}$$

For

$$|(p,h)\rangle = a_h^\dagger(p)|0\rangle, \tag{111}$$

the field-state overlap is

$$\langle 0|A_\mu(x)|(p,h)\rangle = \varepsilon^h_\mu(p)e^{-ip\cdot x}. \tag{112}$$

If the polarization is shifted by

$$\varepsilon^h_\mu(p) \to \varepsilon^h_\mu(p) + c(p)p_\mu, \tag{113}$$

then, after absorbing an overall factor of i into α,

$$\delta A_\mu(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3 2E_p} c(p)p_\mu a_h(p)e^{-ip\cdot x} = \partial_\mu \alpha(x). \tag{114}$$

Hence a change of reference spinor is equivalent to the gauge transformation

$$A_\mu \to A_\mu + \partial_\mu \alpha. \tag{115}$$

2 QED and Non-Abelian Gauge Theories

2.1 Quantum electrodynamics

2.1.1 Vector Field

A relativistic quantum field theory containing a massless spin-one particle can be described using a vector field A_μ with the redundancy

$$A_\mu(x) \to A_\mu(x) + \partial_\mu \alpha(x), \quad \alpha(x) \in \mathbb{R}. \tag{116}$$

The basic gauge-invariant local object is the field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad F_{\mu\nu} = -F_{\nu\mu}. \tag{117}$$

The unique parity-even gauge-invariant kinetic term with two derivatives is

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \tag{118}$$

In four spacetime dimensions,

$$\left[A_\mu\right] = 1, \quad \left[F_{\mu\nu}\right] = 2, \quad [\psi] = \frac{3}{2}. \tag{119}$$

A possible dimension-six contraction of three field strengths vanishes:

$$F_\mu{}^\nu F_\nu{}^\rho F_\rho{}^\mu = 0. \tag{120}$$

The leading pure-photon corrections therefore occur at dimension eight. For example,

$$\Delta\mathcal{L} = \frac{c_1}{\Lambda^4}\left(F_{\mu\nu}F^{\mu\nu}\right)^2 + \frac{c_2}{\Lambda^4}\left(F_{\mu\nu}\tilde{F}^{\mu\nu}\right)^2 + ..., \tag{121}$$

where

$$\tilde{F}^{\mu\nu} = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}. \tag{122}$$

Operators of dimension greater than four are irrelevant in the infrared. Restricting temporarily to operators of dimension at most four gives the free Maxwell theory. Such a theory is conventionally called renormalizable, although higher-dimensional operators are perfectly meaningful within effective field theory.

2.1.2 Coupling photons to matter

Electrons are described by a spin-$\frac{1}{2}$ Dirac field. A manifestly gauge-invariant photon-matter operator is the dipole interaction

$$\Delta\mathcal{L}_{\text{dipole}} = \frac{c}{\Lambda}\overline{\psi}\sigma^{\mu\nu}\psi F_{\mu\nu}, \tag{123}$$

with

$$\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]. \tag{124}$$

This operator has dimension five and is therefore irrelevant. The renormalizable interaction instead couples the vector potential directly to a current:

$$S_{\text{int}} = -\int \mathrm{d}^4x A_\mu(x) J^\mu(x). \tag{125}$$

Under

$$\delta A_\mu = \partial_\mu \alpha, \tag{126}$$

the interaction changes by

$$\delta S_{\rm int} = -\int \mathrm{d}^4x \big(\partial_\mu \alpha\big) J^\mu = \int \mathrm{d}^4x \alpha \partial_\mu J^\mu, \tag{127}$$

where a boundary term was discarded. Gauge invariance therefore requires

$$\partial_\mu J^\mu = 0. \tag{128}$$

This conservation law is automatic when A_μ enters only through the field strength. Indeed, for an antisymmetric tensor $O^{\mu\nu}$,

$$F_{\mu\nu} O^{\mu\nu} = 2\big(\partial_\mu A_\nu\big) O^{\mu\nu} \to -2 A_\nu \partial_\mu O^{\mu\nu}, \tag{129}$$

where the arrow means equality inside the action up to a boundary term. The corresponding current is

$$J^\nu = 2\partial_\mu O^{\mu\nu}, \tag{130}$$

and hence

$$\partial_\nu J^\nu = 2\partial_\nu \partial_\mu O^{\mu\nu} = 0 \tag{131}$$

identically. When A_μ appears directly in the action, however, current conservation becomes a genuine constraint on the matter theory.

The free Dirac Lagrangian is

$$\mathcal{L}_{\rm matter} = i\overline{\psi}\gamma^\mu \partial_\mu \psi - m\overline{\psi}\psi. \tag{132}$$

It is invariant under the global transformation

$$\psi(x) \to e^{-ie\alpha}\psi(x), \quad \overline{\psi}(x) \to \overline{\psi}(x) e^{ie\alpha}, \tag{133}$$

The associated Noether current is

$$j^\mu = -i\frac{\partial \mathcal{L}_{\rm matter}}{\partial \partial_\mu \psi}\psi = \overline{\psi}\gamma^\mu \psi, \tag{134}$$

Coupling this current to the photon field gives the interaction term and if we promote the symmetry to local one, then it is invariant under gauge transformations.

We use the mutually consistent transformations

$$A_\mu(x) \to A_\mu(x) + \partial_\mu \alpha(x), \tag{135}$$

$$\psi(x) \to \psi'(x) = e^{-ie\alpha(x)}\psi(x), \tag{136}$$

$$\overline{\psi}(x) \to \overline{\psi}'(x) = \overline{\psi}(x) e^{ie\alpha(x)}. \tag{137}$$

An ordinary derivative does not transform covariantly. Define the covariant derivative by

$$D_\mu = \partial_\mu + ieA_\mu. \tag{138}$$

Under the joint transformation,

$$D'_\mu \psi'(x) = \big(\partial_\mu + ie\big(A_\mu + \partial_\mu \alpha\big)\big) e^{-ie\alpha}\psi. \tag{139}$$

Expanding the derivative gives

$$D'_\mu \psi'(x) = e^{-ie\alpha}\big(\partial_\mu - ie\partial_\mu\alpha + ieA_\mu + ie\partial_\mu\alpha\big)\psi \tag{140}$$

$$= e^{-ie\alpha} D_\mu \psi. \tag{141}$$

Thus $D_\mu\psi$ transforms in exactly the same way as ψ.

2.1.3 The QED Lagrangian

The gauge-invariant QED Lagrangian is

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\overline{\psi}\gamma^\mu D_\mu\psi - m\overline{\psi}\psi. \tag{142}$$

Expanding the covariant derivative gives

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\overline{\psi}\gamma^\mu\partial_\mu\psi - m\overline{\psi}\psi - eA_\mu\overline{\psi}\gamma^\mu\psi. \tag{143}$$

- This is an Abelian gauge theory with gauge group $U(1)$.
- Gauge invariance forbids a photon mass term

$$\frac{1}{2}m_A^2 A_\mu A^\mu. \tag{144}$$

- The replacement

$$\partial_\mu \to D_\mu \tag{145}$$

  produces the minimal photon-matter interaction.
- The Abelian field strength is linear in A_μ, so the renormalizable Maxwell term contains no photon self-interactions.

Varying the action with respect to A_μ gives

$$\partial_\nu F^{\nu\mu} = J^\mu, \quad J^\mu = e\overline{\psi}\gamma^\mu\psi. \tag{146}$$

These are the inhomogeneous Maxwell equations. Taking the divergence implies current conservation:

$$\partial_\mu J^\mu = \partial_\mu\partial_\nu F^{\nu\mu} = 0. \tag{147}$$

The homogeneous Maxwell equations follow from the Bianchi identity

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0, \tag{148}$$

or equivalently

$$\varepsilon^{\mu\nu\rho\sigma}\partial_\nu F_{\rho\sigma} = 0. \tag{149}$$

Finally, the interaction term

$$\mathcal{L}_{\text{int}} = -eA_\mu\overline{\psi}\gamma^\mu\psi \tag{150}$$

gives the electron-photon vertex

$$-ie\gamma^\mu. \tag{151}$$

2.2 Non-Abelian gauge theories

2.2.1 Motivation

When QED is restricted to operators of dimension at most four, the photon has no self-interactions. Yang-Mills theory generalizes the Abelian gauge principle and describes interacting massless spin-one fields.

The simplest example is based on the group SU(2). Let the matter field transform as a doublet,

$$\psi(x) = \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix}. \tag{152}$$

The generators in the fundamental representation are

$$T^a = \frac{\sigma^a}{2}, \quad [T^a, T^b] = i\varepsilon^{abc}T^c, \quad \mathrm{tr}\big(T^aT^b\big) = \frac{1}{2}\delta^{ab}. \tag{153}$$

For a general compact Lie group G, we write

$$[T^a, T^b] = if^{abc}T^c. \tag{154}$$

The structure constants of SU(2) are

$$f^{abc} = \varepsilon^{abc}. \tag{155}$$

2.2.2 From a global symmetry to a local symmetry

The free matter Lagrangian is

$$\mathcal{L}_0 = i\overline{\psi}\gamma^\mu\partial_\mu\psi - m\overline{\psi}\psi. \tag{156}$$

It is invariant under a global transformation

$$\psi(x) \to \psi'(x) = U\psi(x), \tag{157}$$

$$\overline{\psi}(x) \to \overline{\psi}'(x) = \overline{\psi}(x)U^{-1}, \tag{158}$$

where

$$U = e^{-ig\alpha^aT^a}, \quad \partial_\mu U = 0. \tag{159}$$

If $U = U(x)$ is local, the ordinary derivative produces an extra term:

$$\overline{\psi}'\gamma^\mu\partial_\mu\psi' = \overline{\psi}\gamma^\mu\partial_\mu\psi + \overline{\psi}\gamma^\mu U^{-1}\big(\partial_\mu U\big)\psi. \tag{160}$$

To compensate for this term, introduce a Lie-algebra-valued gauge field

$$A_\mu \equiv A_\mu{}^aT^a \tag{161}$$

and define the covariant derivative

$$D_\mu = \partial_\mu - igA_\mu = \partial_\mu - igA_\mu{}^aT^a. \tag{162}$$

We demand that it transform covariantly:

$$D'_\mu\psi'(x) = U(x)D_\mu\psi(x). \tag{163}$$

2.2.3 Gauge-field transformation

Using $\psi' = U\psi$, covariance requires

$$(\partial_\mu - igA'_\mu)U\psi = U(\partial_\mu - igA_\mu)\psi. \tag{164}$$

Therefore,

$$A'_\mu = UA_\mu U^{-1} - \frac{i}{g}(\partial_\mu U)U^{-1}. \tag{165}$$

This is the finite transformation law of a non-Abelian connection. It contains both a homogeneous rotation and an inhomogeneous derivative term.

For an infinitesimal transformation,

$$U(x) = 1 - ig\alpha^{a(x)}T^a + O(\alpha^2), \tag{166}$$

so that

$$A'_\mu{}^a = A_\mu{}^a - \partial_\mu\alpha^a + gf^{abc}\alpha^b A_\mu{}^c. \tag{167}$$

The homogeneous part shows that the gauge fields transform in the adjoint representation. Unlike the photon, the non-Abelian gauge bosons therefore carry gauge charge.

With the convention used here, the Abelian limit is

$$A_\mu \to A_\mu - \partial_\mu\alpha. \tag{168}$$

Replacing $\alpha \to -\alpha$ gives the alternative QED convention.

2.2.4 Field strength from the covariant derivative

The non-Abelian field strength is defined by

$$[D_\mu, D_\nu] = -igF_{\mu\nu}. \tag{169}$$

Using $D_\mu = \partial_\mu - igA_\mu$ gives

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]. \tag{170}$$

In components,

$$F_{\mu\nu}{}^a = \partial_\mu A_\nu{}^a - \partial_\nu A_\mu{}^a + gf^{abc}A_\mu{}^b A_\nu{}^c. \tag{171}$$

Because

$$D'_\mu = UD_\mu U^{-1}, \tag{172}$$

the field strength transforms homogeneously:

$$F'_{\mu\nu} = UF_{\mu\nu}U^{-1}. \tag{173}$$

Infinitesimally,

$$F'_{\mu\nu}{}^a = F_{\mu\nu}{}^a + gf^{abc}\alpha^b F_{\mu\nu}{}^c. \tag{174}$$

Thus the field strength transforms in the adjoint representation. It is gauge covariant rather than gauge invariant.

2.2.5 The Yang-Mills Lagrangian

The gauge-invariant kinetic term is obtained by tracing over the gauge indices:

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}). \tag{175}$$

For the normalization

$$\mathrm{tr}\big(T^a T^b\big) = \frac{1}{2}\delta^{ab}, \tag{176}$$

this becomes

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4} F_{\mu\nu}{}^{a} F^{\mu\nu a}. \tag{177}$$

Gauge invariance follows from cyclicity of the trace:

$$\mathrm{tr}\big(F'_{\mu\nu} F'^{\mu\nu}\big) = \mathrm{tr}\big(U F_{\mu\nu} U^{-1} U F^{\mu\nu} U^{-1}\big) \tag{178}$$

$$= \mathrm{tr}\big(F_{\mu\nu} F^{\mu\nu}\big). \tag{179}$$

Adding Dirac matter in a representation R gives

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}{}^{a} F^{\mu\nu a} + i\bar{\psi}\gamma^{\mu} D_{\mu}\psi - m\bar{\psi}\psi, \tag{180}$$

with

$$D_{\mu} = \partial_{\mu} - igA_{\mu}{}^{a} T_R^{a}, \tag{181}$$

and

$$\big[T_R^{a}, T_R^{b}\big] = i f^{abc} T_R^{c}. \tag{182}$$

The same coupling g controls both the coupling to matter and the self-interactions of the gauge fields. Gauge invariance does not allow these couplings to be chosen independently.

2.2.6 Gauge-boson self-interactions

Define the linearized field strength

$$f_{\mu\nu}{}^{a} = \partial_{\mu} A_{\nu}{}^{a} - \partial_{\nu} A_{\mu}{}^{a}. \tag{183}$$

Then

$$F_{\mu\nu}{}^{a} = f_{\mu\nu}{}^{a} + g f^{abc} A_{\mu}{}^{b} A_{\nu}{}^{c}. \tag{184}$$

Expanding the Yang-Mills kinetic term gives quadratic, cubic, and quartic terms:

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4} f_{\mu\nu}{}^{a} f^{\mu\nu a} \tag{185}$$

$$-\frac{g}{2} f^{abc} f_{\mu\nu}{}^{a} A^{\mu b} A^{\nu c} \tag{186}$$

$$-\frac{g^2}{4} f^{abc} f^{ade} A_{\mu}{}^{b} A_{\nu}{}^{c} A^{\mu d} A^{\nu e}. \tag{187}$$

The cubic and quartic terms describe three- and four-gauge-boson interactions. Since f^{abc} is antisymmetric, there is no self-coupling involving only one gauge component, but distinct gauge components interact with one another.

2.2.7 General compact gauge groups

For a representation R, the invariant quadratic form is

$$\mathrm{tr}_R\big(T_R^{a} T_R^{b}\big) = T(R)\delta^{ab}. \tag{188}$$

A positive-definite gauge kinetic energy requires a positive invariant inner product on the Lie algebra. This naturally leads to compact gauge groups represented by unitary matrices,

$$U^\dagger U = 1. \tag{189}$$

Important compact possibilities include

- $\mathrm{SU}(N)$,
- $\mathrm{SO}(N)$,
- $\mathrm{Sp}(N)$,
- the exceptional groups G_2, F_4, E_6, E_7, and E_8,
- products of these groups together with Abelian $U(1)$ factors.

For a general group, the SU(2) replacements are

$$T^a \to T^a_R, \quad \varepsilon^{abc} \to f^{abc}. \tag{190}$$

2.2.8 Tree-level Feynman rules

The following rules use the convention

$$D_\mu = \partial_\mu - igA_\mu \tag{191}$$

and Feynman gauge. The gauge-fixing term is

$$\mathcal{L}_{\mathrm{gf}} = -\frac{1}{2}\left(\partial_\mu A^{\mu a}\right)^2. \tag{192}$$

The matter-gauge-boson vertex is

$$ig\gamma^{\mu\alpha}{}_{\beta} T^{a\,i}{}_{j}. \tag{193}$$

The gauge-boson propagator is

$$\frac{-i\eta^{\mu\nu}\delta^{ab}}{p^2 + i\varepsilon}. \tag{194}$$

The fermion propagator is

$$\frac{i\left(\gamma^\rho p_\rho + m\right)^{\alpha}{}_{\beta}}{p^2 - m^2 + i\varepsilon}. \tag{195}$$

For the three-gauge-boson vertex, take all momenta incoming and impose

$$p_1 + p_2 + p_3 = 0. \tag{196}$$

For external legs (a,μ), (b,ν), and (c,ρ), the vertex is

$$V_3 = g f^{abc}\left[\eta^{\mu\nu}(p_2 - p_1)^\rho + \eta^{\nu\rho}(p_3 - p_2)^\mu + \eta^{\rho\mu}(p_1 - p_3)^\nu\right]. \tag{197}$$

For the four-gauge-boson vertex with external legs (a,μ), (b,ν), (c,ρ), and (d,σ), write

$$V_4 = -ig^2(C_1 + C_2 + C_3), \tag{198}$$

where

$$C_1 = f^{abe} f^{cde}(\eta^{\mu\rho}\eta^{\nu\sigma} - \eta^{\mu\sigma}\eta^{\nu\rho}), \tag{199}$$

$$C_2 = f^{ace} f^{bde}(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\sigma}\eta^{\nu\rho}), \tag{200}$$

$$C_3 = f^{ade} f^{bce}(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma}). \tag{201}$$

These rules are sufficient for tree-level calculations involving physical external polarizations. At loop level, a consistent gauge-fixed quantization also requires the Faddeev-Popov ghost sector.

3 Tree-level amplitudes in gauge theory

These notes use the all-incoming convention. Overall phases and signs of amplitudes depend on the normalization of one-particle states and interaction vertices; the spinor-helicity dependence is the essential part.

3.1 Three-particle amplitudes

3.1.1 Complex three-particle kinematics and little-group scaling

For three massless external momenta,

$$p_1 + p_2 + p_3 = 0, \quad p_i^2 = 0. \tag{202}$$

Momentum conservation implies

$$0 = p_1^2 = (-p_2 - p_3)^2 = 2p_2 \cdot p_3, \tag{203}$$

and similarly for the other pairs. In spinor-helicity variables,

$$2p_i \cdot p_j = \langle ij\rangle[ji]. \tag{204}$$

For real momenta, $\langle ij\rangle^* = [ji]$, so all two-particle invariants vanish and the three-point amplitude is kinematically trivial. For complex momenta the angle and square spinors are independent, and there are two possible branches.

Assume, for example, that $\langle 23\rangle \neq 0$. Then

$$\langle 23\rangle[32] = 0 \quad \to \quad [32] = 0. \tag{205}$$

Moreover,

$$0 = \langle 2|\ (p_1 + p_2 + p_3)\ |1] = \langle 23\rangle[31], \tag{206}$$

so $[31] = 0$; similarly $[12] = 0$. Hence

$$[12] = [23] = [31] = 0, \tag{207}$$

and all square spinors are proportional. Conversely, if $[23] \neq 0$, then

$$\langle 12\rangle = \langle 23\rangle = \langle 31\rangle = 0, \tag{208}$$

and all angle spinors are proportional. Thus nonzero three-point amplitudes exist for complex kinematics and are either holomorphic or anti-holomorphic.

Under little-group scaling,

$$|i\rangle \to t_i|i\rangle, \quad |i] \to t_i^{-1}\ |i], \quad \mathcal{A}_n \to t_i^{-2h_i}\mathcal{A}_n. \tag{209}$$

For a holomorphic three-point amplitude, take

$$\mathcal{A}_3\big(1^{h_1}, 2^{h_2}, 3^{h_3}\big) = \langle 12\rangle^{x_{12}}\langle 23\rangle^{x_{23}}\langle 31\rangle^{x_{31}}. \tag{210}$$

Little-group covariance gives

$$-2h_1 = x_{12} + x_{31}, \quad -2h_2 = x_{12} + x_{23}, \quad -2h_3 = x_{23} + x_{31}, \tag{211}$$

and therefore

$$\mathcal{A}_3 = \langle 12\rangle^{h_3 - h_1 - h_2}\langle 23\rangle^{h_1 - h_2 - h_3}\langle 31\rangle^{h_2 - h_3 - h_1}. \tag{212}$$

On the anti-holomorphic branch,

$$\mathcal{A}_3 = [12]^{h_1 + h_2 - h_3}[23]^{-h_1 + h_2 + h_3}[31]^{h_1 - h_2 + h_3}. \tag{213}$$

Thus Lorentz invariance together with little-group scaling fixes the spinor structure of every three-point amplitude, up to coupling constants.

Some examples are

$$\mathcal{A}_3\big(f_1^-, \overline{f}_2^-, \varphi_3\big) = y\langle 12\rangle, \tag{214}$$

and, in QED,

$$\mathcal{A}_3\big(f_1^-, \overline{f}_2^+, \gamma_3^-\big) = -\sqrt{2}e\frac{\langle 13\rangle^2}{\langle 12\rangle}, \tag{215}$$

$$\mathcal{A}_3\big(f_1^-, \overline{f}_2^+, \gamma_3^+\big) = \sqrt{2}e\frac{[23]^2}{[12]}. \tag{216}$$

3.1.2 Three-vector amplitudes

For three massless vectors, the Yang-Mills three-point amplitudes are

$$\mathcal{A}_3(1^-, 2^-, 3^+) = g\frac{\langle 12\rangle^3}{\langle 23\rangle\langle 31\rangle}, \tag{217}$$

$$\mathcal{A}_3(1^-, 2^+, 3^+) = g\frac{[23]^3}{[12][31]}. \tag{218}$$

The first expression is odd under $1 \leftrightarrow 2$. Identical photons therefore have no such self-interaction. For non-Abelian gauge bosons the antisymmetry is carried by the color factor,

$$\mathcal{A}_3(1_a^-, 2_b^-, 3_c^+) = gf^{abc}\frac{\langle 12\rangle^3}{\langle 23\rangle\langle 31\rangle}, \quad f^{abc} = -f^{bac}. \tag{219}$$

Higher-dimension or nonlocal interactions can generate other little-group-compatible structures. For example,

$$\mathcal{A}_3(1^-, 2^-, 3^-) = \frac{1}{\Lambda^2}\langle 12\rangle\langle 23\rangle\langle 31\rangle, \tag{220}$$

which can arise from a dimension-six F^3 operator, while a nonlocal interaction may give

$$\mathcal{A}_3(1^-, 2^+, 3^+) = g'\frac{\langle 13\rangle\langle 12\rangle}{\langle 23\rangle^3}. \tag{221}$$

3.2 BCFW recursion

3.2.1 General complex deformation and factorization

Higher-point tree amplitudes can be reconstructed from lower-point on-shell amplitudes. Introduce a complex deformation of the external momenta,

$$\hat{p}_{i(z)} = p_i + zr_i, \tag{222}$$

with

$$r_i \cdot r_j = 0 \quad \text{for all } i, j, \quad p_i \cdot r_i = 0, \quad \sum_i r_i = 0. \tag{223}$$

These conditions preserve the on-shell constraints and total momentum conservation:

$$\hat{p}_i^2 = 0, \quad \sum_i \hat{p}_i = \sum_i p_i. \tag{224}$$

For a factorization channel I, define

$$P_I = \sum_{i \in I} p_i, \quad R_I = \sum_{i \in I} r_i, \quad \hat{P}_{I(z)} = P_I + z R_I. \tag{225}$$

The shifted propagator goes on shell at

$$\hat{P}_I^2(z) = P_I^2 + 2z P_I \cdot R_I = -P_I^2 \frac{z - z_I}{z_I}, \quad z_I = -\frac{P_I^2}{2 P_I \cdot R_I}. \tag{226}$$

The shifted tree amplitude $\hat{\mathcal{A}}_{n(z)}$ is a rational function of z. Cauchy's theorem gives

$$\mathcal{A}_n = -\sum_{z_I} \mathrm{Res}_{z=z_I} \frac{\hat{\mathcal{A}}_{n(z)}}{z} + B_n, \tag{227}$$

where B_n is the contribution from $z = \infty$. Near a physical pole, tree-level factorization implies

$$\hat{\mathcal{A}}_{n(z)} \to \hat{\mathcal{A}}_{L(z_I)} \frac{1}{\hat{P}_I^2(z)} \hat{\mathcal{A}}_{R(z_I)}, \tag{228}$$

and

$$\frac{1}{\hat{P}_I^2(z)} = -\frac{z_I}{z - z_I} \frac{1}{P_I^2}. \tag{229}$$

If the deformation is chosen so that $B_n = 0$, the BCFW recursion relation is

$$\mathcal{A}_n = \sum_{I,h} \hat{\mathcal{A}}_{L(z_I)} \frac{1}{P_I^2} \hat{\mathcal{A}}_{R(z_I)}. \tag{230}$$

The sum runs over factorization channels and over the helicity and species of the internal on-shell state. The subamplitudes on both sides are evaluated at $z = z_I$.

3.2.2 The two-line gauge-theory shift

Choose two external legs i and j and perform the $[i, j\rangle$ shift

$$|\hat{i}] = |i] + z\,|j], \quad |\hat{j}\rangle = |j\rangle - z|i\rangle, \tag{231}$$

while

$$|\hat{i}\rangle = |i\rangle, \quad |\hat{j}] = |j]. \tag{232}$$

Equivalently,

$$\hat{p}_i = p_i + z|i\rangle[j|, \quad \hat{p}_j = p_j - z|i\rangle[j|. \tag{233}$$

For a negative-helicity gauge boson on leg i and a positive-helicity gauge boson on leg j, both polarization vectors scale as z^{-1}. The part of a tree diagram connecting the two shifted legs grows at worst as z, hence

$$\hat{\mathcal{A}}_{n(z)} = O(z^{-1}) \quad \text{as} \quad z \to \infty, \tag{234}$$

so $B_n = 0$. A more refined argument shows that pure-gluon amplitudes also admit additional valid shifts.

3.3 Four-point amplitudes and gauge-algebra consistency

3.3.1 Compton scattering

Consider the QED amplitude

$$\mathcal{A}_4\big(f_1^-, \overline{f}_2^+, \gamma_3^-, \gamma_4^+\big) \tag{235}$$

and use the $[3,4\rangle$ shift

$$|\hat{3}] = |3] + z\,|4], \quad \big|\hat{4}\big\rangle = |4\rangle - z|3\rangle. \tag{236}$$

The first factorization channel has internal momentum $\hat{P} = p_2 + \hat{p}_4$. Its pole is fixed by

$$\hat{P}^2 = \big\langle 2\hat{4}\big\rangle [42] = 0, \quad z_{24} = \frac{\langle 24\rangle}{\langle 23\rangle}. \tag{237}$$

At the pole,

$$\big\langle 1\hat{4}\big\rangle = \langle 14\rangle - z_{24}\langle 13\rangle = \frac{\langle 14\rangle\langle 23\rangle - \langle 13\rangle\langle 24\rangle}{\langle 23\rangle} = -\frac{\langle 12\rangle\langle 34\rangle}{\langle 23\rangle}. \tag{238}$$

The two BCFW residues are

$$\mathcal{A}_{24} = 2e^2 \frac{\langle 13\rangle^2\langle 23\rangle}{\langle 24\rangle\langle 12\rangle\langle 34\rangle}, \tag{239}$$

$$\mathcal{A}_{14} = -2e^2 \frac{\langle 13\rangle^3}{\langle 14\rangle\langle 12\rangle\langle 34\rangle}. \tag{240}$$

Adding them,

$$\mathcal{A}_4 = -2e^2 \frac{\langle 13\rangle^2}{\langle 12\rangle\langle 34\rangle}\left(\frac{\langle 13\rangle}{\langle 14\rangle} - \frac{\langle 23\rangle}{\langle 24\rangle}\right). \tag{241}$$

Using the Schouten identity,

$$\frac{\langle 13\rangle}{\langle 14\rangle} - \frac{\langle 23\rangle}{\langle 24\rangle} = \frac{\langle 12\rangle\langle 34\rangle}{\langle 14\rangle\langle 24\rangle}, \tag{242}$$

one obtains

$$\mathcal{A}_4\big(f_1^-, \overline{f}_2^+, \gamma_3^-, \gamma_4^+\big) = -2e^2 \frac{\langle 13\rangle^2}{\langle 14\rangle\langle 24\rangle}. \tag{243}$$

A different valid shift can produce the same result from a single on-shell diagram. Equality of different BCFW representations is a useful consistency condition.

3.3.2 Adding color: the Lie algebra from on-shell consistency

Let the fermion-fermion-vector three-point amplitude be

$$\mathcal{A}_3\big(f_i^-, \overline{f}_j^+, g_A^-\big) = T^A{}_{ij} \frac{\langle 13\rangle^2}{\langle 12\rangle}. \tag{244}$$

Computing the colored Compton amplitude from the two fermion-exchange channels gives

$$\mathcal{A}_4 = \frac{\langle 13\rangle^2}{\langle 12\rangle\langle 34\rangle}\left(T^A{}_{ik}T^B{}_{kj}\frac{\langle 23\rangle}{\langle 24\rangle} - T^B{}_{ik}T^A{}_{kj}\frac{\langle 13\rangle}{\langle 14\rangle}\right). \tag{245}$$

With another shift, one also exposes the gauge-boson exchange channel:

$$\mathcal{A}_4 = \frac{\langle 13\rangle^2}{\langle 24\rangle\langle 41\rangle} T^A{}_{ik}T^B{}_{kj} - f_{ABC}T^C{}_{ij}\frac{\langle 13\rangle^3}{\langle 12\rangle\langle 34\rangle\langle 41\rangle}. \tag{246}$$

Requiring both constructions to agree forces the couplings to close under a Lie algebra. With Hermitian generators,

$$T^A{}_{ik}T^B{}_{kj} - T^B{}_{ik}T^A{}_{kj} = if^{ABC}T^C{}_{ij}, \tag{247}$$

or equivalently

$$\left[T^A, T^B\right] = if^{ABC}T^C. \tag{248}$$

If the factor i is absorbed into the definition of the cubic coupling, the same relation is written without the explicit i. Thus locality, factorization, and shift independence already imply the Lie-algebra structure of the gauge couplings.

3.4 Vanishing sectors, MHV amplitudes, and Parke-Taylor

3.4.1 Helicity selection and color ordering

At tree level, pure Yang-Mills amplitudes with all equal helicities vanish. The same is true when only one gluon has the opposite helicity:

$$A_{n(1^-,2^-,\ldots,n^-)} = 0, \quad A_{n(1^+,2^-,\ldots,n^-)} = 0. \tag{249}$$

Parity gives the corresponding all-plus and one-minus statements. These results follow inductively from BCFW recursion because every possible factorization contains a vanishing lower-point amplitude.

The first nonzero generic tree amplitudes have two negative-helicity gluons and are called MHV amplitudes:

$$A_{n(1^-,2^-,3^+,\ldots,n^+)}. \tag{250}$$

The full Yang-Mills amplitude can be decomposed into color-ordered partial amplitudes,

$$\mathcal{A}_{n(1^{a_1},\ldots,n^{a_n})} = g^{n-2}\sum_{\sigma\in\frac{S_n}{Z_n}} A_{n(\sigma(1),\ldots,\sigma(n))}\ \mathrm{tr}(T^{a_{\sigma(1)}}\ldots T^{a_{\sigma(n)}}). \tag{251}$$

The partial amplitude A_n is associated with a fixed cyclic ordering. Its physical poles occur only when adjacent ordered momenta become collinear or when sums of cyclically adjacent momenta go on shell.

3.4.2 Four-point MHV amplitude and the n-point formula

For the ordering $(1,2,3,4)$, BCFW recursion gives, in the overall sign convention of the handwritten notes,

$$A_4(1^-,2^-,3^+,4^+) = -g^2\frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle\langle 34\rangle\langle 41\rangle}. \tag{252}$$

The denominator displays poles only for adjacent particles in the color ordering, while the numerator is fixed by little-group scaling. Suppressing the overall coupling and convention-dependent phase suggests the n-point ansatz

$$A_{n(1^-,2^-,3^+,\ldots,n^+)} = \frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle\ldots\langle n-1,n\rangle\langle n1\rangle}. \tag{253}$$

This formula follows by induction. Apply the $[1,n\rangle$ shift,

$$|\hat{1}] = |1] + z\,|n], \quad |\hat{n}\rangle = |n\rangle - z|1\rangle. \tag{254}$$

For the chosen ordering and helicities, only the channel

$$\hat{P} = p_{n-1} + \hat{p}_n \tag{255}$$

contributes. The on-shell condition gives

$$\langle n-1,\hat{n}\rangle = 0, \quad z_* = \frac{\langle n-1,n\rangle}{\langle n-1,1\rangle}. \tag{256}$$

At this pole, the Schouten identity gives

$$\langle n-2,\hat{n}\rangle = \frac{\langle n-2,n\rangle\langle n-1,1\rangle - \langle n-2,1\rangle\langle n-1,n\rangle}{\langle n-1,1\rangle} = \frac{\langle n-2,n-1\rangle\langle 1n\rangle}{\langle n-1,1\rangle}. \tag{257}$$

Substituting the induction hypothesis for the left subamplitude and the appropriate three-point amplitude on the right yields

$$A_{n(1^-,2^-,3^+,\ldots,n^+)} = \frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle\ldots\langle n-1,n\rangle\langle n1\rangle}. \tag{258}$$

More generally, if the two negative-helicity legs are r and s,

$$A_{n(1^+,\ldots,r^-,\ldots,s^-,\ldots,n^+)} = \frac{\langle rs\rangle^4}{\prod_{a=1}^{n}\langle a,a+1\rangle}, \quad n+1\equiv 1. \tag{259}$$

This is the Parke-Taylor formula for color-ordered tree-level gluon amplitudes.

Amplitudes with three negative helicities are called NMHV; more generally, amplitudes with $k+2$ negative-helicity gluons are N^k MHV. BCFW recursion expresses each sector in terms of lower-point amplitudes. In QCD this classification is fundamentally a tree-level organization; loop amplitudes have a richer structure.

Bibliography